

Adaptive Trial Design for the RAS-Inhibitor Era: Combinations, Resistance, and Biomarker-Guided Therapy

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Today's argument

New RAS therapies need new kinds of trials.

Methods to optimize combination doses are established. Designs that adapt to emerging resistance are enrolling.

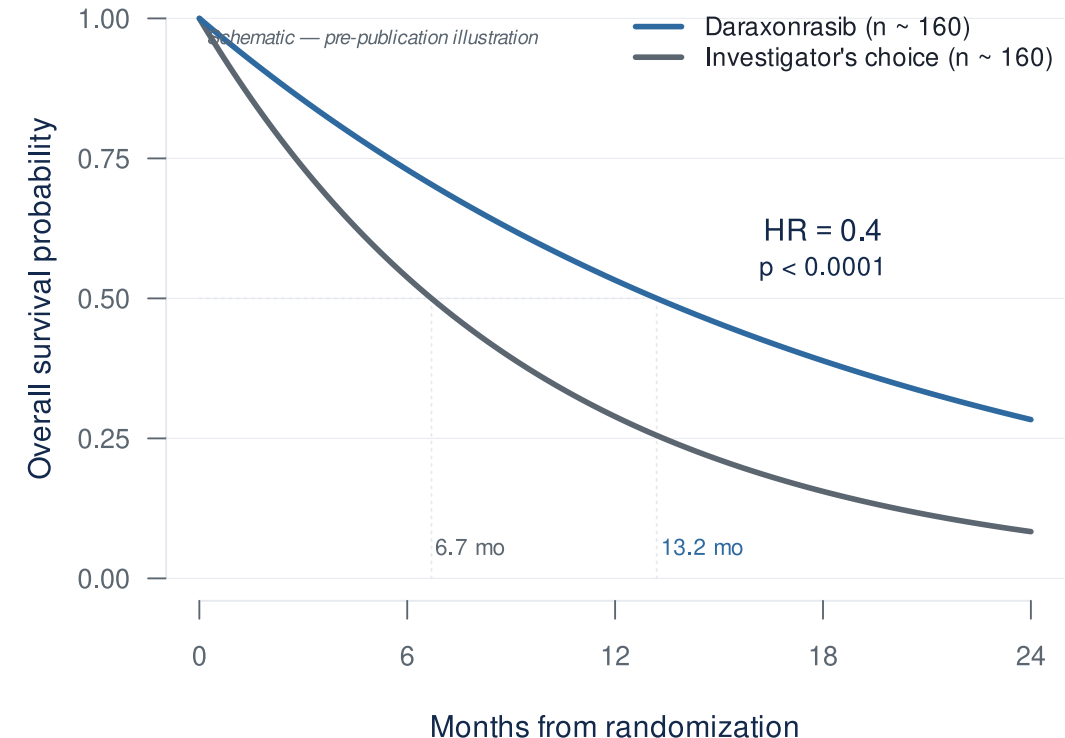
The difficulty: getting them calibrated and launched.

The clinical case

Why combinations, and why now?

The RAS revolution has arrived for PDAC

- RASolute 302 (2L mPDAC, *NEJM* 2026)¹:
 - **Median OS: 13.2 vs 6.7 months**
 - **HR 0.40 — 60% reduction in risk of death ($p < 0.0001$)**
- Daraxonrasib (RMC-6236) — oral, multi-selective RAS(ON) inhibitor
- ASCO 2026 Plenary, May 31 (Wolpin)



...but monotherapy has visible ceilings

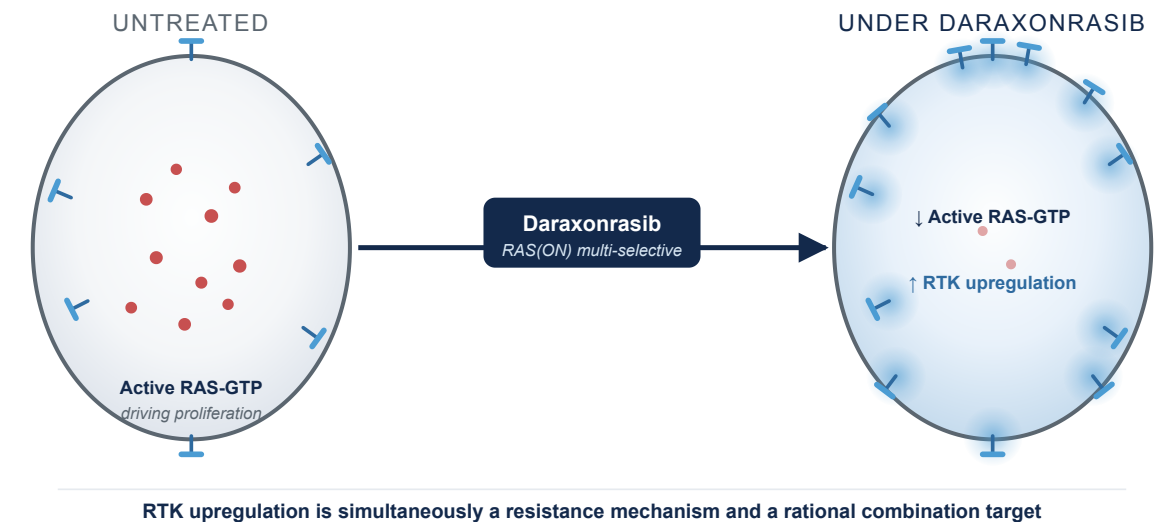
RMC-6236-001 (PDAC cohort, NEJM 2026)²



Half of patients on monotherapy at 300 mg already need dose adjustments. Combinations layer on top — with less room to spare.

Biomarker-driven combinations are the next move

- Daraxonrasib drives **RTK upregulation** on PDAC cell surface — the mechanistic case for biomarker-driven, resistance-preemptive combinations³
- **1L combination cohort** (daraxonrasib 200 mg + GnP, n=40)⁴:
 - ORR: **58%** · DCR: **90%** · 6-mo PFS: **84%** · 6-mo OS: **90%**



The dose-finding problem

Why MTD only may not be best for RAS combinations, and what replaces it.

Why combinations break the rules you grew up on

- **Rivière's review of 162 combination Phase I trials: 88% used 3+3** even when both drugs were escalated^{5,6}.
- **Why 3+3 doesn't fit combinations.** Built for single-agent cytotoxics⁷, it assumes monotonic dose-toxicity, Cycle-1-observable DLTs, and a single dose dimension — combinations strain all three.
- **No joint-surface model.** In practice: pick A's dose, fix it, add B. Two monotherapy escalations run in series; the two-drug surface is never characterized.

Joint surface = the two-drug benefit-risk landscape — how efficacy and toxicity respond at every (A, B) dose pair, not just along one drug's axis.

The Amgen case: sotorasib + pembrolizumab

Same sponsor, two different designs:

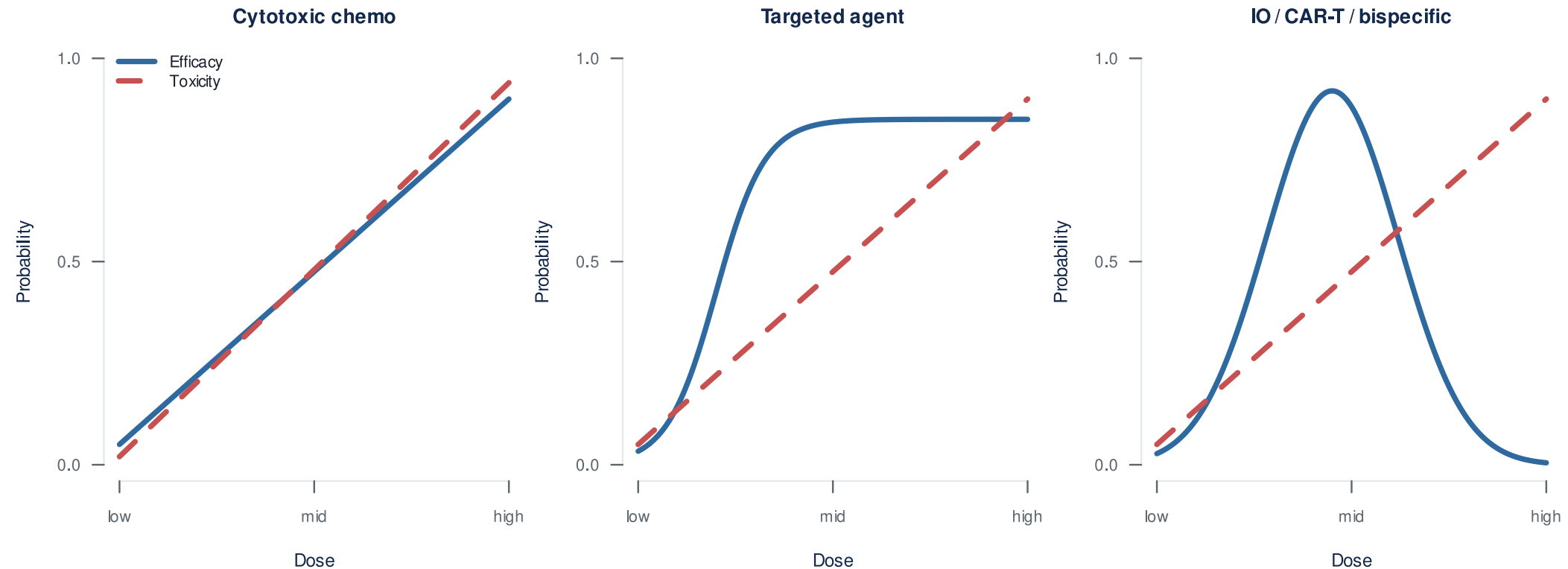
- **Monotherapy (CodeBreakK 100):** Bayesian model-based, single-agent surface
- **Combination:** IO dose fixed, only sotorasib varied — joint surface never modeled

The defining toxicity:

- Severe **immune-mediated** hepatotoxicity, not predictable from the single agent
- **88%** of Grade 3-4 events outside the Cycle 1 DLT window⁸

What MTD-only dose-finding was never built to handle, all at once: joint surface unsearched, non-additive toxicity, late-onset toxicity.

Why MTD-only dose-finding doesn't fit targeted agents and immunotherapies



The cost of MTD-only dose-finding

- **Sotorasib** (KRAS G12C, first-in-class, approved 2021)¹⁰:
 - 960 mg selected as the **highest dose tested**, no DLTs — a more-is-better default, not an MTD
 - FDA required a **240 mg vs 960 mg** comparison. **PFS was similar** at one-quarter the exposure¹¹
- **MTD $\neq$ optimal dose** for targeted agents and immunotherapies
- **More than 40% of patients** in small-molecule registrational trials require dose reductions or interruptions¹²

Same drug class as daraxonrasib.

Project Optimus: the regulatory ground has shifted

How OBD selection actually works

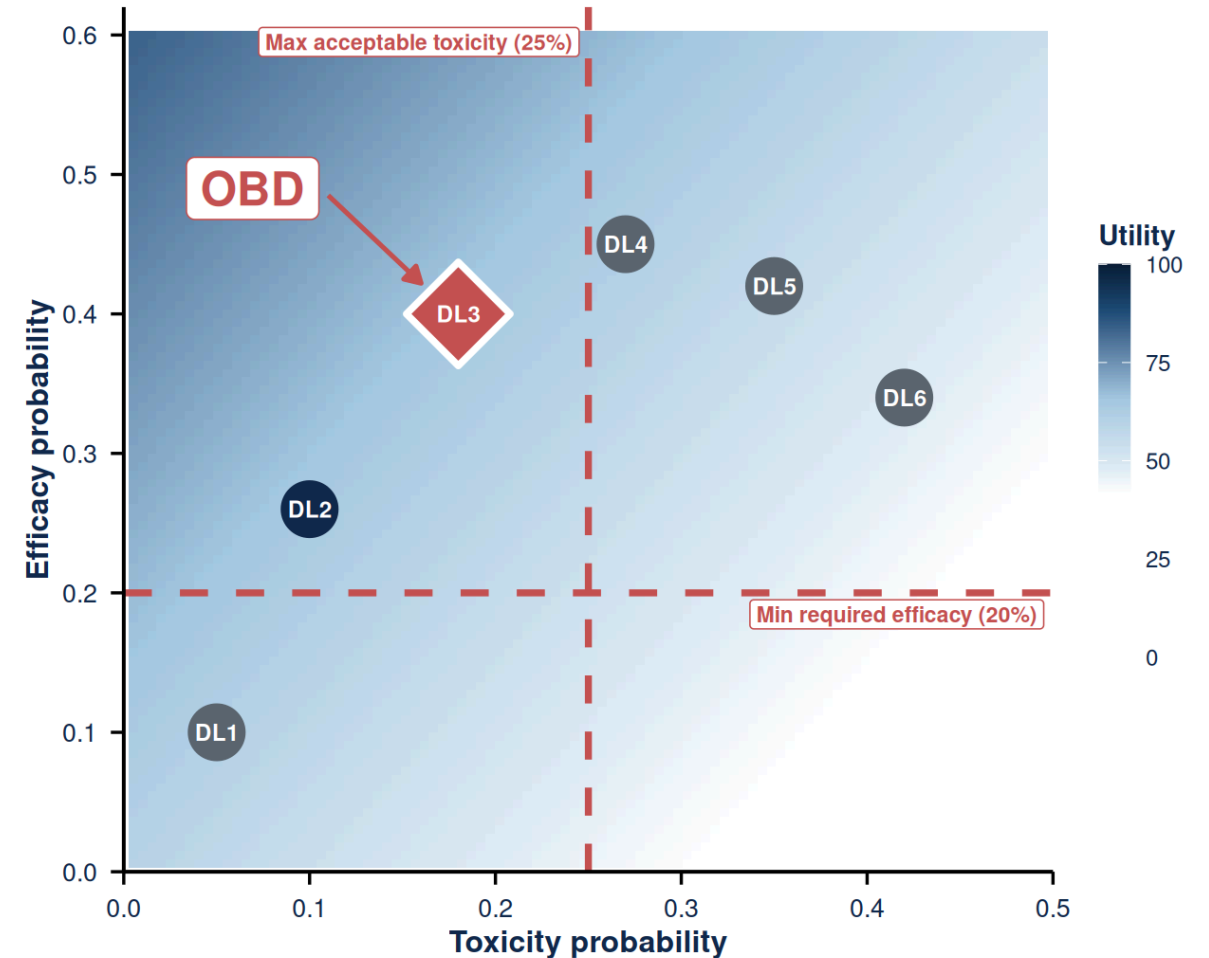
Utility-based dose-finding, in one sentence.

“A response is worth this; a Grade 3 toxicity costs this.”

The team writes that utility table down up front. Bayesian, model-assisted design then searches inside the safe envelope for the dose that **maximizes benefit-risk**, not toxicity.

Elicited utility table (Zhou 2019 framework)¹⁴:

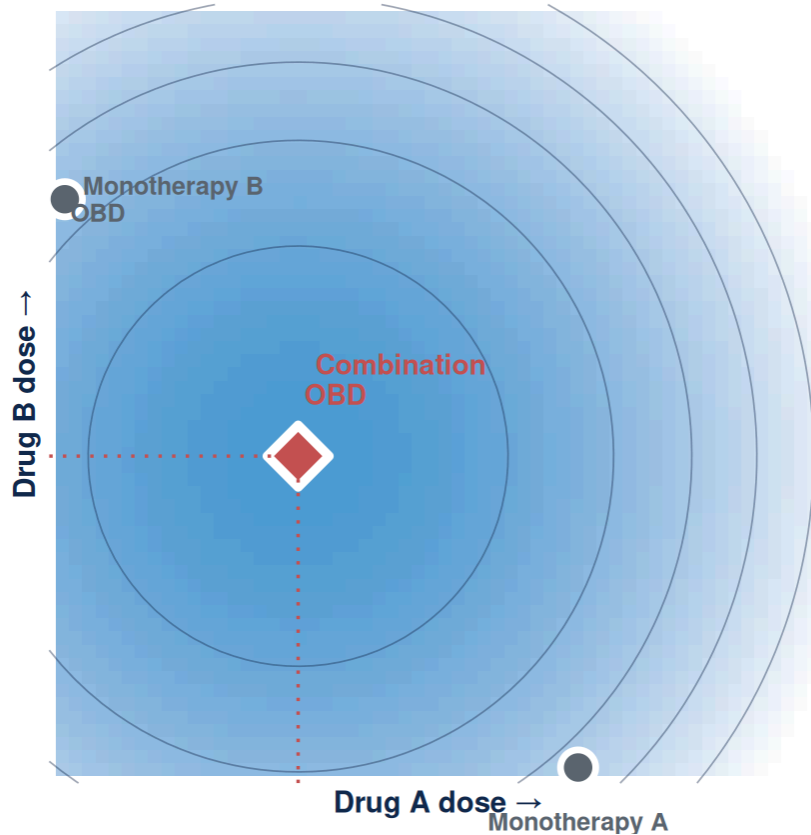
	TOX	NO TOX
Response	40	100
No response	0	60



OBD = admissible dose (inside red lines) with the highest utility.

Combination dose selection in RAS is harder

Combinations make MTD-only dose-finding worse on every axis.



- The combination OBD can sit **below both single-agent OBDs**
- Single-agent data **cannot reconstruct** the joint surface
- For RAS, the **toxicity margin is already spent** at the monotherapy dose

Fix what you know. Search what you don't.

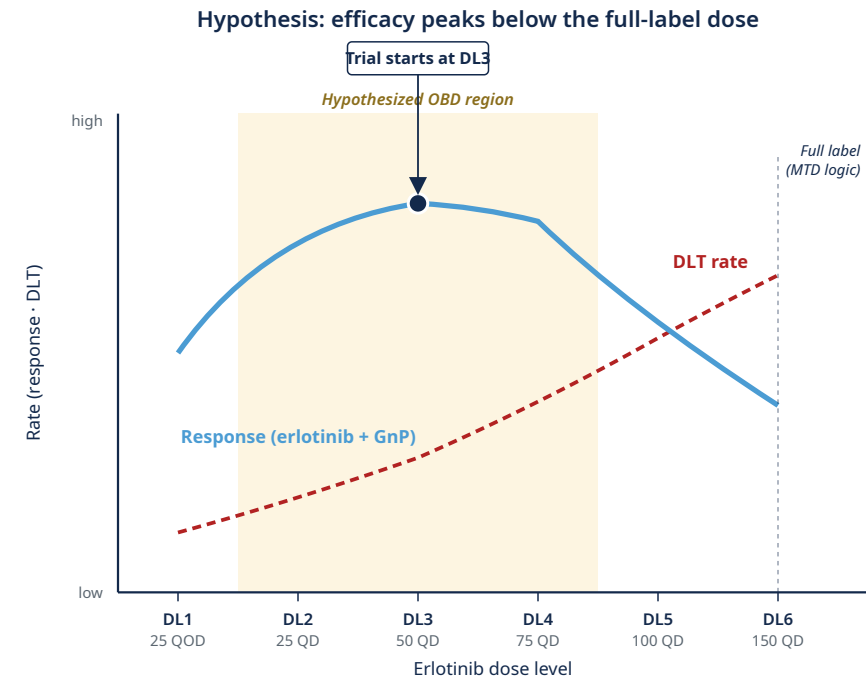
RMC-GI-102 (daraxonrasib + GnP) fixed daraxonrasib at 200 mg without searching the joint surface⁴.

Conceptual schematic of the benefit-risk surface. The combination OBD sits below both single-agent OBDs; single-agent data alone cannot reconstruct the contour.

Finding the right erlotinib dose in basal-like PDAC

- **Trial:** LCCC2220-DCT (Somasundaram, PI), basal-like PDAC by PurIST¹⁷
- **Combination:** erlotinib + biweekly gemcitabine/nab-paclitaxel
- **Hypothesis:** lower-dose erlotinib may be more efficacious (OBD-below-MTD)¹⁸
- **Six dose levels:** 25 mg QOD → 25 / 50 / 75 / 100 / 150 mg QD; **start at DL3**

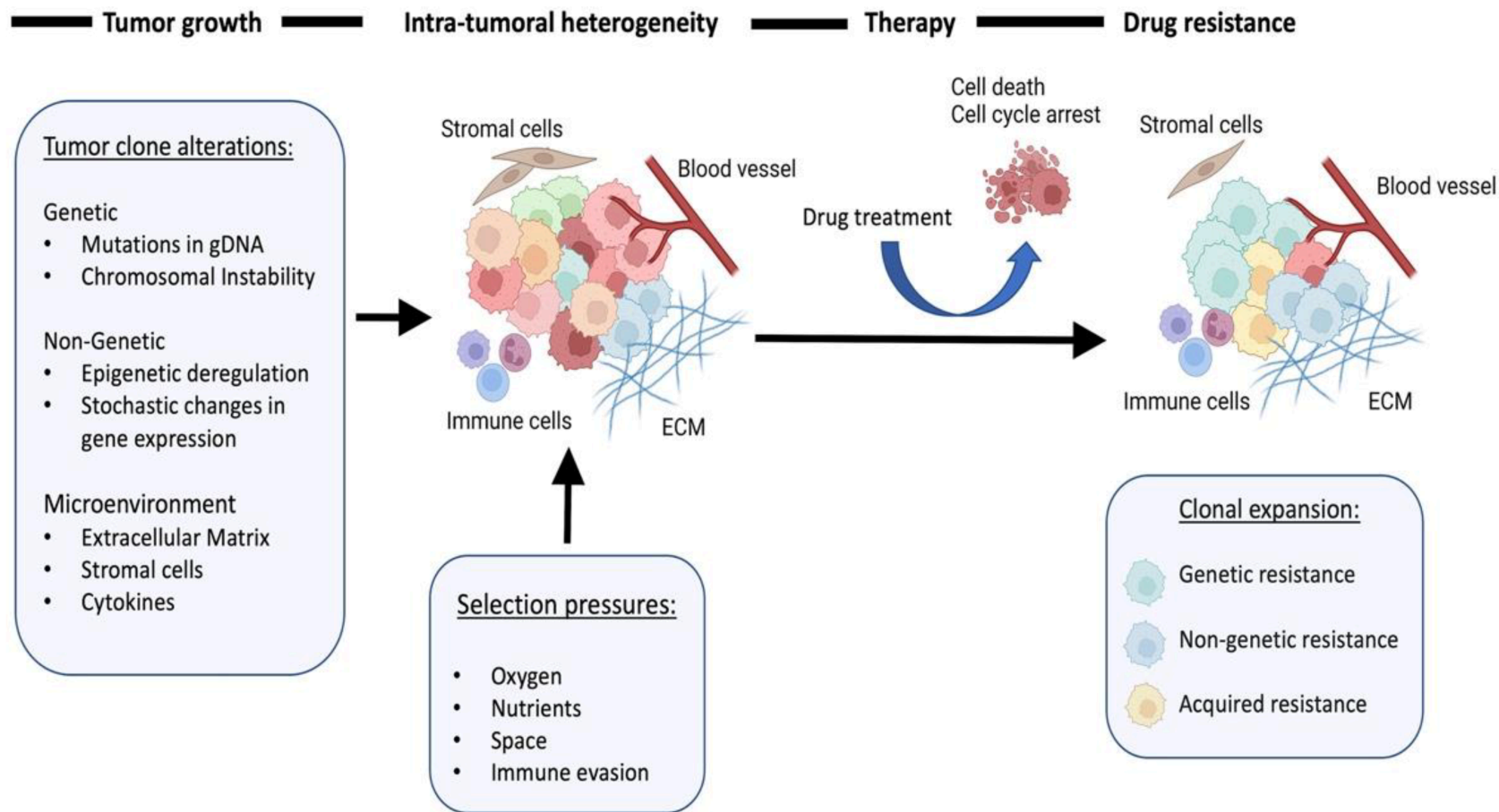
Design: *Stage 1* — establish the safety envelope¹⁹. *Stage 2* — find the dose with the best response/toxicity trade-off¹⁴. **Up to 52 patients; ~28 expected at the OBD.**



Resistance is the next problem

Phase 2 onward: each patient is a moving target

Resistance is a dynamic process

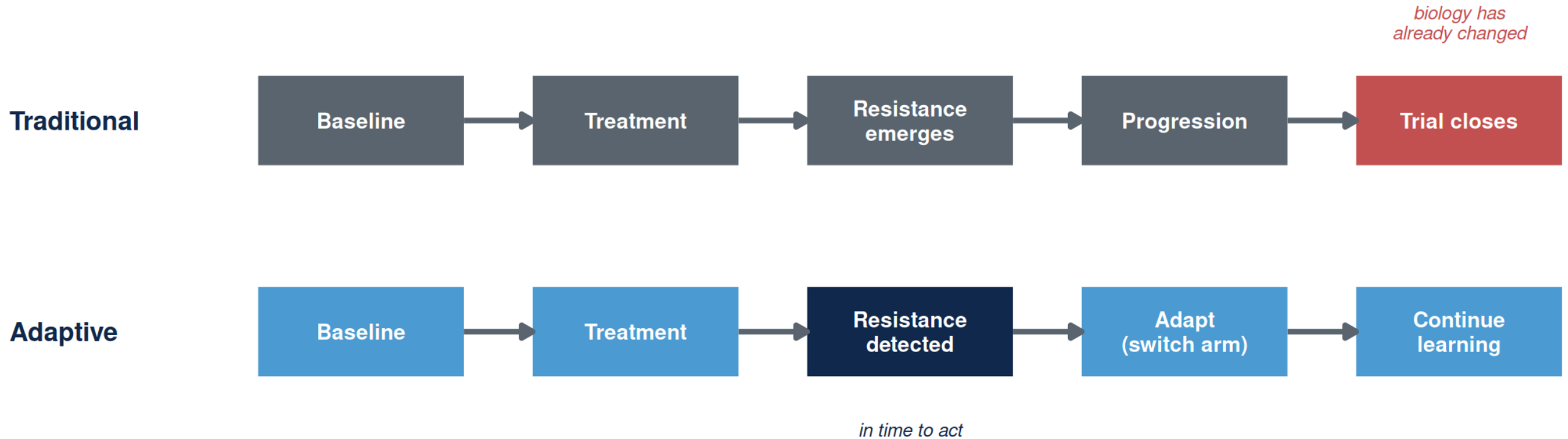


Resistance mechanisms emerge under RAS-inhibitor pressure

Daraxonrasib monotherapy in mPDAC (n=44, paired ctDNA): 59% show RAS-pathway resistance at progression — KRAS amplification in 36%²¹

Adapted from Karagiannis & Rampias, Cancers 2022²⁰.

Why classical trial designs miss resistance



Genomic → ctDNA · non-genomic / plasticity → biopsies, transcriptomics, imaging. A resistance-aware trial has to do **both** during enrollment^{22,23}.

ADAPT / EVOLVE: what a resistance-aware trial looks like

ARPA-H's resistance-tracking template.

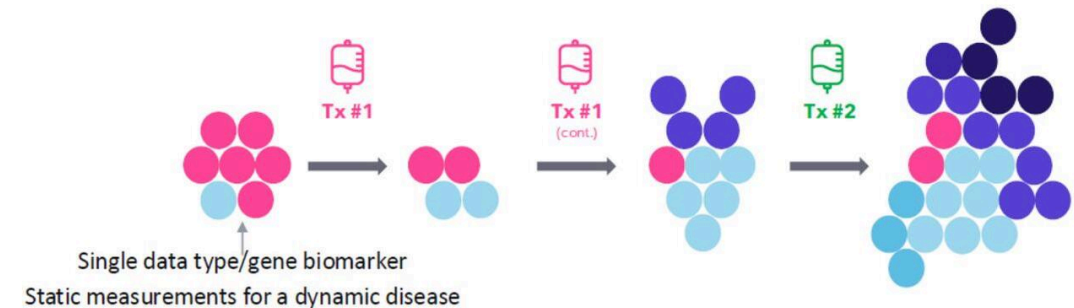
- Longitudinal resistance detection
- Biomarker-guided adaptation
- Adaptive arm-swaps on emerging biology

ARPA-H ADAPT → EVOLVE (UNC-led; Carey MPI, Rashid MPI).
Enrolling.

ADAPT Goal: Identify and target changes in metastatic cancers

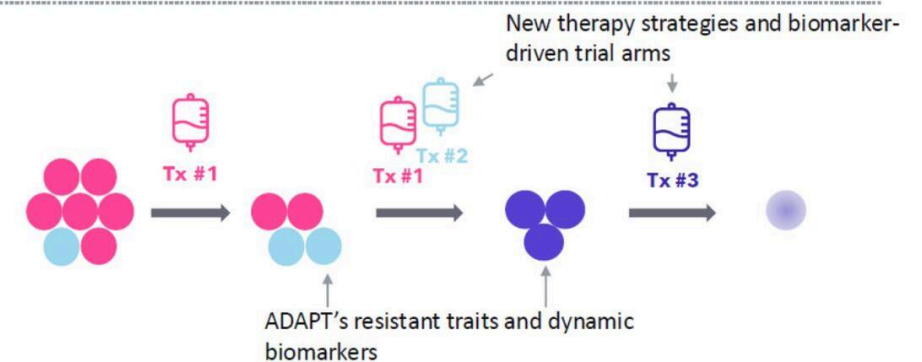
Current State

- Cancer changes during treatment. Therapies **not** often informed by tumor change to cancer resistance traits.
- There are only **limited** tumor measurement biomarker analyses, which are most often performed **prior to** treatment.



ADAPT's upcoming "future state"

- Therapies **will be** informed by and tailored to observed cancer resistance traits using biomarkers.
- **In-depth** tumor measurements will be taken **before, during and after** treatment.

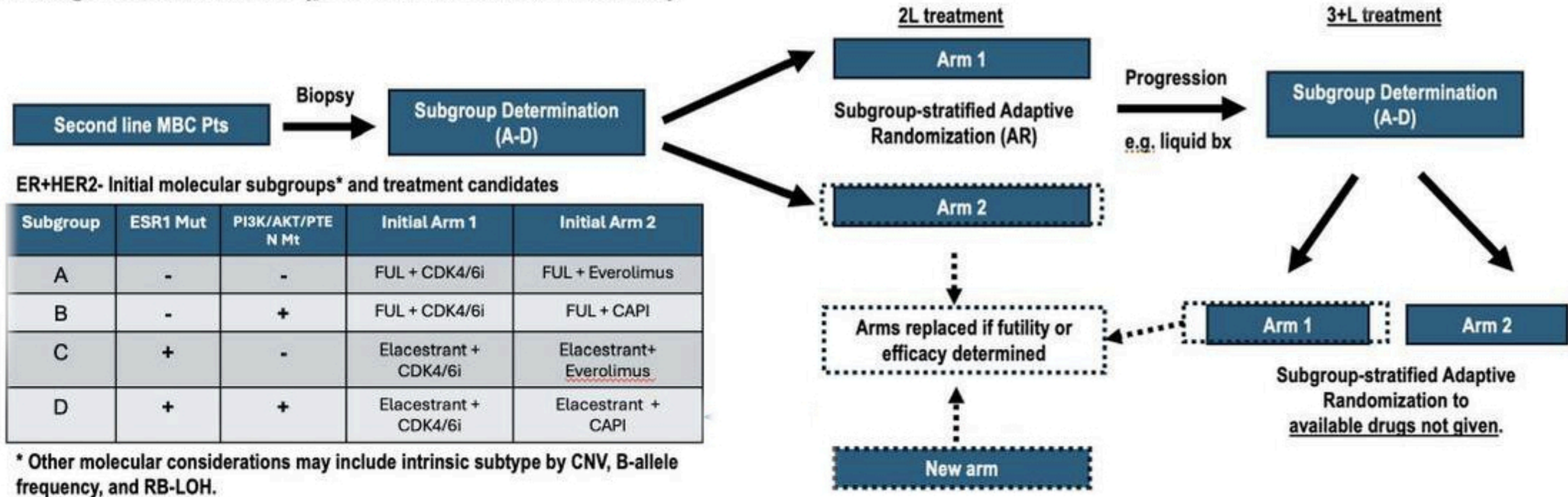


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ARPA-H ADAPT — current state (top) vs dynamic-biomarker state (bottom).

EVOLVE-BDT — and what this would look like in PDAC

First generation trials (pre-TA1 novel biomarkers)



TBCRC EVOLVE-BDT flow — biomarker-stratified subtrials, adaptive randomization, in-trial arm replacement on futility or efficacy.

EVOLVE-BDT (real, enrolling)

ER+/HER2– and TNBC, biomarker-stratified subtrials with Bayesian arm-swap on futility/efficacy.

PDAC analog (hypothetical)

Basal-like vs classical by PurlST¹⁷, same Bayesian arm-swap logic on daraxonrasib-combination arms.

Same architecture, different disease.

What this demands of the statistical design

Four design axes any such trial needs.

Adaptive stopping

Bayesian futility /
efficacy boundaries
at each interim

Real-time biomarker discovery

Pre-specified signatures
+ agnostic
discovery window

Longitudinal monitoring

Serial ctDNA,
imaging, biopsies
on every patient

In-trial evaluation

Biomarker-guided
cohorts within the
same protocol

Methods are ready. We're operationalizing them in PDAC trials — starting with PANGAEA.

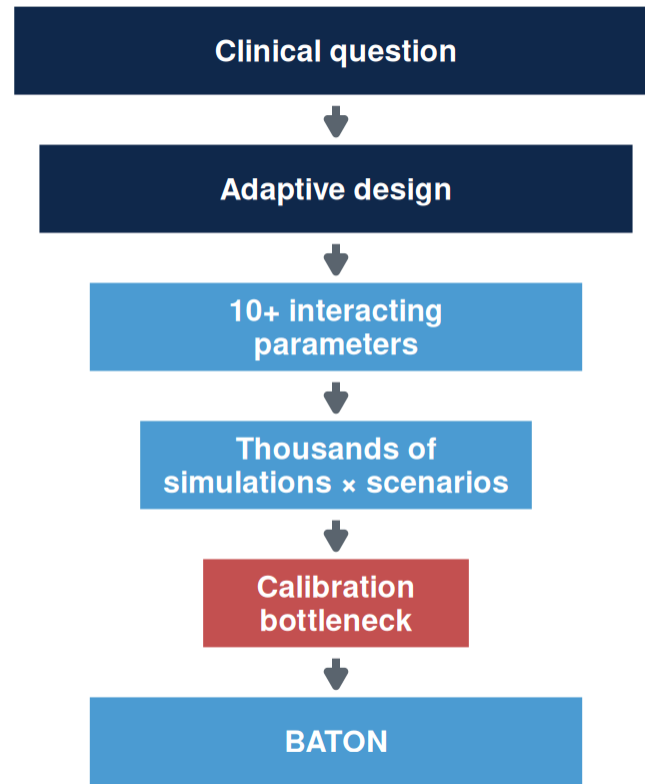
Putting these into practice

Why don't we see more of these trials?

Funding, accrual, and biopsy logistics are real.

But the barrier that quietly kills good designs — and the one we can actually remove — is calibration.

*Calibration tunes a design to hit its **operating characteristics** — the trial's report card: how often it catches real drug effects and rejects non-working drugs, how many patients enrolled, how long it runs.*



Bayesian designs adopted 75%

Dose-optimization plans 30%

Methods adopted. Dose-optimization plans — where calibration cost bites — not²⁴.

Our group: several months per design. For combined dose + resistance designs, hand-tuning is infeasible.

The calibration challenge: four competing goals

Designing an adaptive trial requires achieving all four simultaneously.

1. FDA compliance

Power ≥ 0.80
Type I ≤ 0.10

2. Minimize patients

Smaller $E[N]$
Efficient use of
scarce patients

3. Stop bad early

Futility stopping
Protect patients
from inactive arms

4. Accelerate good

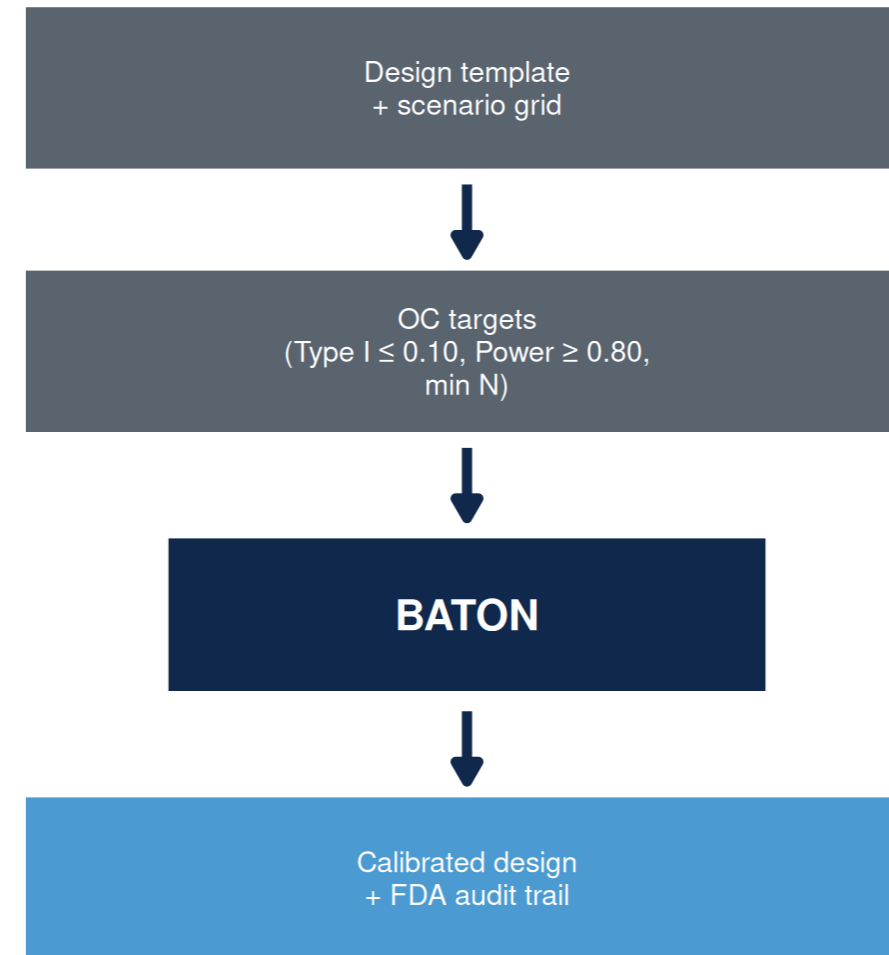
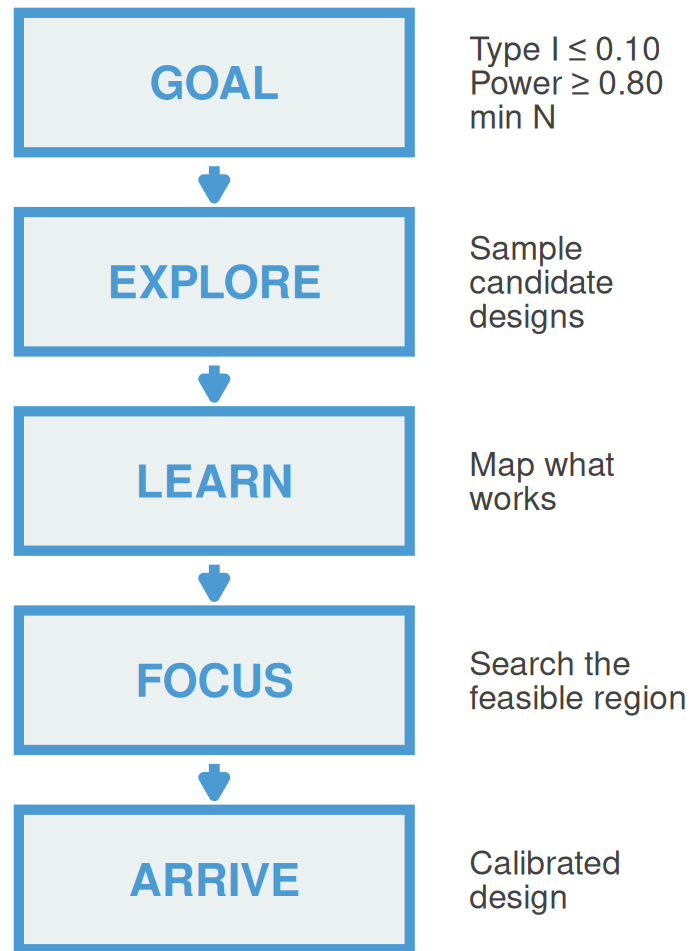
Efficacy stopping
Faster access
when drug works

***“We could design the science.
We could not design the trial.”***

BATON: the calibration tool

BATON = Bayesian Adaptive Trial Optimization²⁵

BATON turns clinically necessary trial complexity into calibrated, launchable designs.



You bring the clinical inputs; BATON returns the calibration.

You bring the clinical design; BATON evaluates hundreds of candidates in silico and returns the ones that meet your targets. R package: github.com/naimurashid/BATON²⁵

A BATON-calibrated design hits all four goals — for PANGEA Phase II

1. FDA compliance ✓

Type I 7.0% (target $\leq 10\%$)
Power 82.2% (target $\geq 80\%$)

2. Minimize patients ✓

~23 patients spared
(avg N 53 vs Max N 76)

3. Stop bad early ✓

78% early futility stop
when drug is inactive

4. Accelerate good ✓

81% early efficacy stop
when drug works

70% fewer evaluations than grid search — same optimum. Manual calibration: weeks to months. BATON: hours, with an FDA-grade audit trail.

What this means for a PDAC patient

Mr. Hernandez, 62, mPDAC, basal-like by PurlST, progressed on 1L GnP.

Traditional fixed Phase II

- All **76 patients** enrolled before any look
- Drug effect discovered only at study end
- **~36+ months** to read out

PANGEA Phase II (BATON-calibrated)

- **78% chance** of early futility stop **when the drug is inactive**
- Average patients enrolled: **~53** if inactive / **~58** if active
- **27.3 months** expected duration

Sooner answer, fewer patients on a losing arm — without losing power.

PANGEA Phase II: calibrated by BATON

- After Phase I OBD → **Phase II Bayesian adaptive RCT**: GnP + erlotinib (at OBD) vs GnP alone, primary PFS
- **Design assumes** reference 5.5 mo → 9.2 mo (HR 0.58)²⁶
- **BATON-calibrated** four parameters under Type I ≤ 0.10 , Power ≥ 0.80 ²⁵

Achieved operating characteristics:

METRIC	VALUE
Power	82.2%
Type I error	7.0%
Max N	76 (82 accrual w/ buffer)
Expected N under H_0	53.0
Expected N under H_1	58.1
Pr(early futility stop H_0)	77.9%
Pr(early efficacy stop H_1)	81.4%
Expected duration	27.3 mo

H_0 = drug inactive (no PFS benefit over GnP alone) · H_1 = drug active at the assumed effect size (HR 0.58)

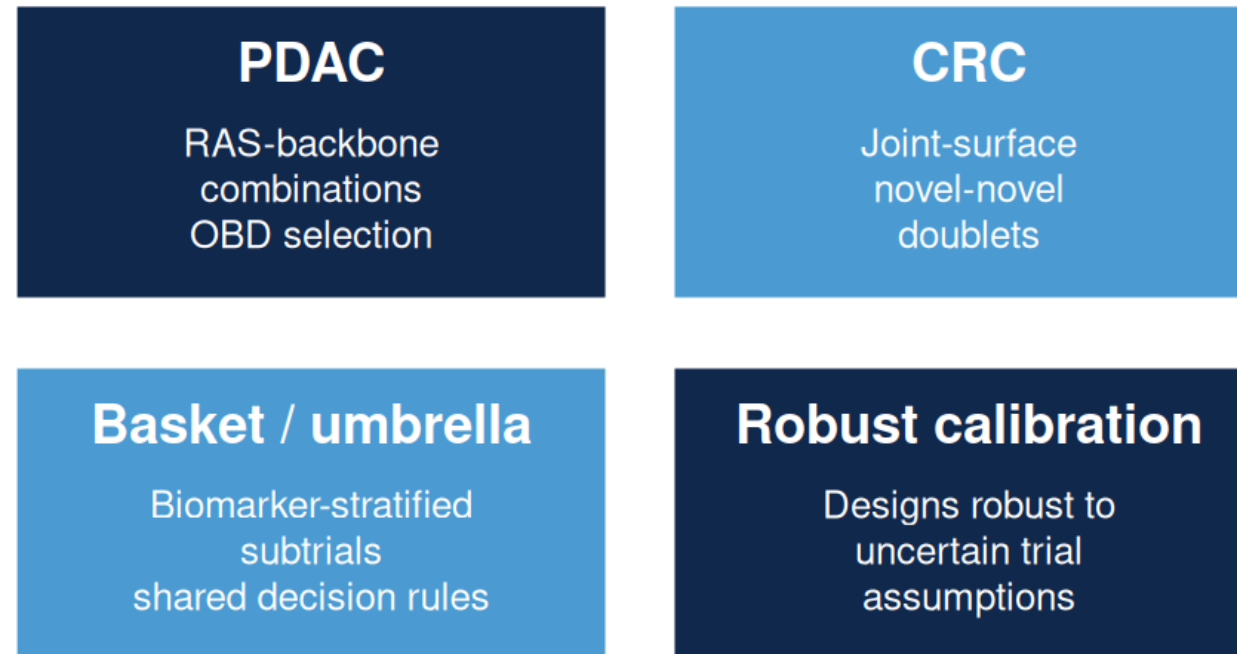
Read as: 78% chance of stopping early when the drug is inactive; 81% chance of stopping early when it works.

Without BATON: four interacting parameters rarely hit both constraints simultaneously. **With BATON:** joint calibration against both constraints, ~200 evaluations.

***An unexpected finding:** caps set too low suppress early-efficacy declarations — even when the drug works, the trial waits for the final analysis to say so. BATON returned the **best trade-off curve** across candidate caps; we chose 76 to preserve early-efficacy stopping. Manual calibration would have anchored on the first feasible design.*

Where this is going

BATON calibrates **aggressive**, **conservative**, or **balanced** designs — and ports across GI applications:



BATON is the infrastructure that makes adaptive trial design routine across the GI portfolio.

Take-home

1. **The RAS era needs new kinds of trials** — biomarker-guided, dose-optimized, resistance-aware.
2. **The methods exist.** Bayesian adaptive designs handle all three.
3. **BATON makes them practical to calibrate and launch.**

Acknowledgments

Clinical collaborators

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- Ashwin Somasundaram (PANGAEA PI)

EVOLVE MPI team

- Charles Perou (UNC) · Ian Krop, Eric Winer (Yale) · Antonio Wolff (Hopkins)

Methods collaborators

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- NCI U01-CA274298

References

1. Eileen M. O'Reilly, Zev A. Wainberg, Andrew E. Hendifar, Mitesh J. Borad, Filippo Pietrantonio, Shubham Pant, Pascal Hammel, Chiara Cremolini, Gulam A. Manji, Paul E. Oberstein, Ignacio Garrido-Laguna, Christoph Springfeld, Nilofer S. Azad, Makoto Ueno, Stephen Y. Chui, Ying Zhang, Hina Patel, Yeonju Lee, Zeena Salman, Brian M. Wolpin, RASolute 302 Trial Investigators. Daraxonrasib or chemotherapy in previously treated metastatic pancreatic cancer. New England Journal of Medicine. 2026. doi:[10.1056/NEJMoa2605555](https://doi.org/10.1056/NEJMoa2605555)
2. Brian M. Wolpin, others. Daraxonrasib in previously treated advanced RAS-mutated pancreatic cancer. New England Journal of Medicine. 2026;394(18):1790–802. doi:[10.1056/NEJMoa2505783](https://doi.org/10.1056/NEJMoa2505783)
3. Yongxian Zhuang, others. The RAS(ON) multi-selective inhibitor daraxonrasib (RMC-6236) induces receptor tyrosine kinase cell-surface expression on pancreatic cancer cells. Cancer Research. 2026;86(5_Suppl_1):PR009. doi:[10.1158/1538-7445.RASONCOTHER26-PR009](https://doi.org/10.1158/1538-7445.RASONCOTHER26-PR009)
4. Brian M. Wolpin, Benjamin L. Musher, Gulam A. Manji, others. Daraxonrasib plus chemotherapy as first-line treatment for patients with metastatic pancreatic adenocarcinoma. AACR Annual Meeting 2026, Abstr LB407 (RMC-GI-102, NCT06445062); 2026.
5. Marie-Karelle Rivière, Christophe Le Tourneau, Xavier Paoletti, Frédéric Dubois, Sarah Zohar. Designs of drug-combination phase I trials in oncology: A systematic review of the literature. Annals of Oncology. 2015;26(4):669–74. doi:[10.1093/annonc/mdu516](https://doi.org/10.1093/annonc/mdu516)

6. Jennifer A. Harrington, Graham M. Wheeler, Michael J. Sweeting, Adrian P. Mander, Duncan I. Jodrell. Adaptive designs for dual-agent phase I dose-escalation studies. *Nature Reviews Clinical Oncology*. 2013;10(5):277–88. doi:[10.1038/nrclinonc.2013.35](https://doi.org/10.1038/nrclinonc.2013.35)
7. Christophe Le Tourneau, J. Jack Lee, Lillian L. Siu. Dose escalation methods in phase I cancer clinical trials. *Journal of the National Cancer Institute*. 2009;101(10):708–20. doi:[10.1093/jnci/djp079](https://doi.org/10.1093/jnci/djp079)
8. Bob T. Li, Gerald S. Falchook, Gregory A. Durm, others. OA03.06 CodeBreak 100/101: First report of safety/efficacy of sotorasib in combination with pembrolizumab or atezolizumab in advanced KRAS p.G12C NSCLC. *Journal of Thoracic Oncology*. 2022;17(9):S10–1.
9. Amita Patnaik, others. Phase I study of pembrolizumab (MK-3475; anti-PD-1 monoclonal antibody) in patients with advanced solid tumors. *Clinical Cancer Research*. 2015;21(19):4286–93. doi:[10.1158/1078-0432.CCR-14-2607](https://doi.org/10.1158/1078-0432.CCR-14-2607)
10. David S. Hong, others. KRAS^{G12C} inhibition with sotorasib in advanced solid tumors. *New England Journal of Medicine*. 2020;383(13):1207–17. doi:[10.1056/NEJMoa1917239](https://doi.org/10.1056/NEJMoa1917239)
11. Maximilian J. Hochmair, Karim Vermaelen, Giannis Mountzios, others. Sotorasib (960 mg or 240 mg) once daily in patients with previously treated KRAS G12C-mutated advanced NSCLC. *European Journal of Cancer*. 2024;208:114204. doi:[10.1016/j.ejca.2024.114204](https://doi.org/10.1016/j.ejca.2024.114204)
12. Hao Zhu, Alex N. Phipps, Ying Yuan, Brad A. Davidson, Stacy S. Shord, Jiang Liu, Patricia M. LoRusso. FDA–AACR strategies for optimizing dosages for oncology drug products: Selecting dosages for first-in-human trials. *Clinical Cancer Research*. 2025;31(23):4872–81. doi:[10.1158/1078-0432.CCR-25-0095](https://doi.org/10.1158/1078-0432.CCR-25-0095)

13. U.S. Food and Drug Administration. Optimizing the dosage of human prescription drugs and biological products for the treatment of oncologic diseases. Guidance for Industry (final); 2024.
14. Yanhong Zhou, J. Jack Lee, Ying Yuan. A utility-based bayesian optimal interval (U-BOIN) phase I/II design to identify the optimal biological dose for targeted and immune therapies. *Statistics in Medicine*. 2019;38(28):5299–316.
15. Ruitao Lin, Yanhong Zhou, Fangrong Yan, Daniel Li, Ying Yuan. BOIN12: Bayesian optimal interval phase I/II trial design for utility-based dose finding in immunotherapy and targeted therapies. *JCO Precision Oncology*. 2020;4:1393–402.
16. Peter F. Thall, Elizabeth Garrett-Mayer, Nolan A. Wages, Susan Halabi, Ying Kuen Cheung. Current issues in dose-finding designs: A response to the US FDA oncology center of excellence project optimus. *Clinical Trials*. 2024;21(3):267–72.
17. Naim U. Rashid, Xianlu L. Peng, Chong Jin, Richard A. Moffitt, Keith E. Volmar, Brian A. Belt, Roheena Z. Panni, Timothy M. Nywening, Stephen G. Herrera, Kathleen J. Moore, Sharon G. Hennessey, Ashley B. Morrison, Ryan Kawalerski, Apoorve Nayyar, Albert E. Chang, Bradley Schmidt, Hong Jin Kim, David C. Linehan, Jen Jen Yeh. Purity independent subtyping of tumors (PurIST): A clinically robust, single-sample classifier for tumor subtyping in pancreatic cancer. *Clinical Cancer Research*. 2020;26(1):82–92. doi:[10.1158/1078-0432.CCR-19-1467](https://doi.org/10.1158/1078-0432.CCR-19-1467)
18. Steven J. Cohen, others. A phase 1b study of erlotinib in combination with gemcitabine and nab-paclitaxel in patients with previously untreated advanced pancreatic cancer. *Cancer Chemotherapy and Pharmacology*. 2016;77(4):693–701.

19. Suyu Liu, Ying Yuan. Bayesian optimal interval designs for phase I clinical trials. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*. 2015;64(3):507–23.
20. Dimitris Karagiannis, Theodoros Rampias. Cancer evolution in precision medicine era. *Cancers (Basel)*. 2022;14(8):1885. doi:[10.3390/cancers14081885](https://doi.org/10.3390/cancers14081885)
21. Ida Aronchik, others. Mechanisms of resistance to the RAS(ON) multi-selective inhibitor daraxonrasib in pancreatic cancer. ENA Symposium 2025, Revolution Medicines poster; 2025.
22. Mark M. Awad, others. Acquired resistance to KRAS G12C inhibition in cancer. *New England Journal of Medicine*. 2021;384(25):2382–93. doi:[10.1056/NEJMoa2105281](https://doi.org/10.1056/NEJMoa2105281)
23. others. Mechanisms of resistance to oncogenic KRAS inhibition in pancreatic cancer. *Cancer Discovery*. 2024;14(11):2135–61. doi:[10.1158/2159-8290.CD-24-0177](https://doi.org/10.1158/2159-8290.CD-24-0177)
24. Deepak Bhamidipati, others. Impact of FDA project optimus guidance on the design of early-phase clinical trials. *JCO Oncology Advances*. 2025. doi:[10.1200/OA-25-00084](https://doi.org/10.1200/OA-25-00084)
25. Amber M. Young, others. BATON: Constrained Bayesian optimization for calibrating adaptive trials. 2026.
26. Daniel D. Von Hoff, others. Increased survival in pancreatic cancer with nab-paclitaxel plus gemcitabine. *New England Journal of Medicine*. 2013;369(18):1691–703. doi:[10.1056/NEJMoa1304369](https://doi.org/10.1056/NEJMoa1304369)
27. Channing J. Der, Jen Jen Yeh. Advances in RAS therapeutics for pancreatic cancer. *New England Journal of Medicine*. 2026;394(18):1857–61. doi:[10.1056/NEJMe2600517](https://doi.org/10.1056/NEJMe2600517)

Questions